

Answer Key — Biology Grade 11

Sub-Strand 1.2: Ecology • Lesson 6: How Many Fish Are Left? Capture-Recapture Simulation to Estimate Population Size

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	A	A	B	C	B	A

USE AFTER: PREDICT PHASE — Predict: individual prediction and think-pair-share on how to find out how many fish are in the river

Q1. Fishers say there are fewer fish in a Kenyan river, but the fish are hidden under the water and keep moving. Why do scientists use a sampling method such as capture-recapture instead of counting every fish?

- A. Fishers are not allowed to count fish
- B. Sampling gives the exact number of fish, but counting does not
- C. Counting every moving fish in a river is not practical, so a sample is used to estimate the total**
- D. Fish are too small to be counted by people

Correct answer: C. Mobile organisms hidden in water cannot be counted one by one, so a marked sample is used to estimate the population. The result is an estimate, not an exact count.

USE AFTER: OBSERVE PHASE — Observe: capture-recapture simulation with 100 counters (Recorder/Calculator role)

Q2. In a capture-recapture simulation, 15 counters are captured and marked, then returned and mixed. A second sample of 20 counters is taken and 5 of them are marked. What is the estimated population?

- A. 60**
- B. 15
- C. 100
- D. 75

Correct answer: A. Estimate = (number marked x number in second sample) / number of marked recaptured = $(15 \times 20) / 5 = 300 / 5 = 60$.

USE AFTER: OBSERVE PHASE — Observe: Recapturer mixes and takes the second random sample

Q3. In the simulation, a group forgets to mix the marked counters thoroughly with the others before taking the second sample. Why is this a problem?

- A. The second sample may not represent the whole population, so the estimate can be badly wrong**
- B. The container would hold more than 100 counters
- C. The marked counters would change colour
- D. The formula would no longer have a number for the marked counters

Correct answer: A. The method assumes the marked individuals are fully mixed. If they are not, the second sample is not a fair sample of the population and the estimate becomes unreliable.

USE AFTER: EXPLAIN PHASE — Explain: Lake Victoria problem comparing the second year (50 tagged, 60 caught, 12 tagged)

Q4. Fisheries officers on Lake Victoria catch and tag 50 Nile tilapia and release them. Later they catch 60 fish, and 12 of them have tags. What is the estimated number of tilapia in the lake?

- A. 600
- B. 250**
- C. 500
- D. 10

Correct answer: B. Estimate = $(50 \times 60) / 12 = 3000 / 12 = 250$.

USE AFTER: EXPLAIN PHASE — Explain: discussion of what happens to the estimate if more or fewer marked individuals are recaptured

Q5. Fisheries officers tag 40 fish in a lake. In Year A, they catch 50 fish and 5 are tagged. In Year B, they again catch 50 fish and 10 are tagged. What does this suggest about the fish population in Year B compared with Year A?

- A. It is the same, because the same number of fish were tagged and caught
- B. It cannot be compared, because the estimates are not numbers
- C. It is smaller, because a larger share of the catch is tagged**
- D. It is larger, because more tagged fish were found

Correct answer: C. Year A estimate = $(40 \times 50) / 5 = 400$. Year B estimate = $(40 \times 50) / 10 = 200$. More tagged fish recaptured means the marked fish make up a larger share of the population, so the population is smaller.

USE AFTER: EXPLAIN PHASE — Explain: listing the assumptions of the method (marks are not lost) and how they might be broken

Q6. During a capture-recapture study, many of the tags fall off the tagged fish before the second catch. How will this affect the estimated population size?

- A. There will be no effect, because the formula uses only the second sample
- B. The estimate will be too large, because fewer tagged fish are recognised in the second catch**
- C. The estimate will be exact, because tags do not matter
- D. The estimate will be too small, because fewer fish are counted

Correct answer: B. If tags are lost, fewer marked fish are recaptured. Since the number recaptured is in the denominator, the estimate becomes larger than the true population.

USE AFTER: MODEL BUILDING PHASE — Model: revising the class river model with an arrow and a question mark for the causes of the decline (500 to 250)

Q7. A class uses capture-recapture on a river and gets an estimate of 500 fish in one year and 250 fish in the next year. Which conclusion is best supported by this evidence?

- A. The estimates suggest the fish population has fallen, but more investigation is needed to find the cause**
- B. The estimates prove that overfishing is the cause of the decline
- C. The estimates are exact counts, so exactly 250 fish died
- D. The estimates prove that pollution is the cause of the decline

Correct answer: A. The method estimates numbers and shows a likely decline, but it does not explain why the numbers dropped. Causes such as pollution or overfishing need further evidence.